



What is SmartStock?

Managing inventory manually is a slow, error-prone process that leads to stockouts, overstocking, and wasted resources. SmartStock is an AI-powered inventory and demand forecasting system built for any stock-heavy environment, from warehouses and retail to pharmaceutical stores.

Two machine learning models work in tandem: Meta's Prophet forecasts future demand volumes over user-defined horizons, while a Random Forest classifier flags items at near-term risk of running out. Built on FastAPI, PostgreSQL, and Docker, with a CustomTkinter desktop GUI, SmartStock turns reactive stock management into a proactive, data-driven process.

How does it works?

1. **Scan** — Staff scan items using a USB barcode reader to instantly look up or update stock levels

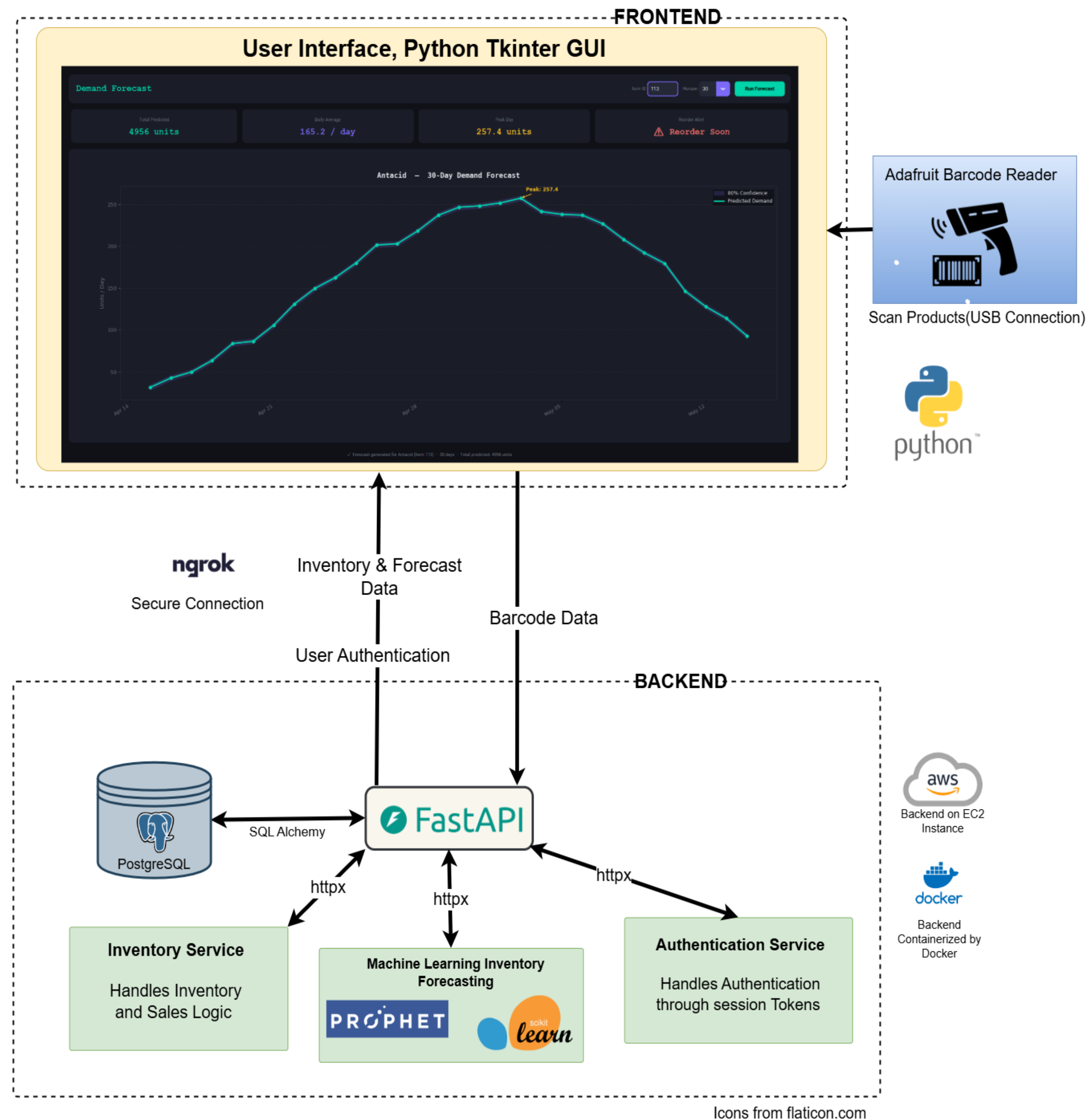
2. **Track** — Every stock movement is logged automatically, building a full audit trail over time.

3. **Forecast** — Meta's Prophet analyses movement history to predict future demand over 7, 14, or 30 days.

4. **Assess** — A Random Forest classifier evaluates each item and flags those at near-term stockout risk.

5. **Act** — Managers use the desktop GUI to review forecasts, monitor stock levels, and make restocking decisions.

Architecture Diagram



Tools & Technologies



Python — Primary programming language used across the backend, ML models, and desktop GUI



FastAPI — Powers the REST API and microservices architecture for inventory and forecasting



PostgreSQL — Relational database storing all inventory items and stock movement history



Docker & Docker Compose — Containerises the full application stack for consistent, portable deployment



Meta Prophet — Time-series forecasting model that predicts future stock demand from historical movement data



scikit-learn — Provides the Random Forest classifier used to assess near-term stockout risk



CustomTkinter — Builds the desktop GUI, giving users an accessible interface for stock management and forecasting

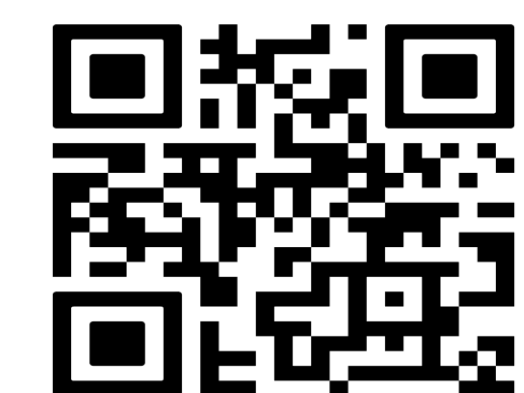


AWS — Cloud platform for hosting and deploying the application infrastructure

Summary of Results

SmartStock successfully delivers a fully functional inventory management system with AI-driven forecasting capabilities. The system supports real-time stock tracking, barcode scanning, and automated movement logging across any stock-heavy environment. Meta's Prophet model generates accurate demand forecasts over 7, 14, and 30-day horizons, while the Random Forest classifier reliably identifies items at near-term stockout risk. The FastAPI backend, containerised with Docker, exposes a clean REST API consumed by both the desktop GUI and the forecasting microservice.

GitHub



Visit <https://github.com/anesuleo/>